UM-1: an accurate approximate solution of the coupled field equations

dirac16complex project (with Claude Opus 5.5)

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Version 1.2 of 2026-10-17 (1.1: sections 4.2, 5.2 and 7 corrected after the analysis `ANALYSIS.md`; 1.2: sections 4.2, 6 and 7 for UM-1 version 1.3, `GAPS.md`; the numbers of the solution are unchanged), step [4] of the author's order of 2026-10-17, with the author's correction: "use Kohn-Sham theory for the dirac16complex field, which couples to all other fields". The model is defined in `MODEL.md`, its field equations are derived in `FIELD_EQUATIONS.md`, and the origin of every ingredient is in `PROVENANCE.md`. The scripts are in `solution/` and every number in this document comes from them. The tables of section 9 are generated by `solution/solve_model.py` from the outputs of the run, and the checks are in `solution/reports/solution-checks.json`. The labels are those of `MODEL.md`: DERIVED, TAKEN, STANDARD, ASSUMED, INVENTED, OPEN. A change to the model that the solution forces is labelled CHANGED.

Coordinates are the author's: $X = \{x_0, x_1, \ldots, x_7\}$. x_0 is the hidden space-like direction, $z = 6Hx_0$ is real and $y = \ln|\sin z|/(6H) \in [-L, 0]$ (with $(\sin^2 z)^{1/6}$ on the positive branch); x_1, x_2, x_3 are 3-space, x_4 is the time and x_5, x_6, x_7 are the extra times. The block (x_0, x_5, x_6, x_7) has the scale factor $f = e^{-a_4} b_4^{1/2}$ and the rate $h_4 = \dot{f}/f$. Here $k = e^{2a_4}/b_4 = 1/f^2$, $A = \dot{a}_4$ is the four-dimensional Hubble rate, and the Lovelock weights are $\alpha_k = w_k H^{2-2k}$.

1. Summary of the results

1. The solution is a hierarchy of approximations in small parameters (section 2): the bookkeeping parameter ϵ of the metric series, $(A/H)^2$ (the late-time expansion rate against the cell scale, 4×10^{-93} today), $\kappa\rho/H^2$ (the sources against the cell curvature) and the portal amplitudes. At each order the field equations become ordinary equations or eigenproblems, which are solved numerically with stated errors.

2. Gravity (section 3, DERIVED, `solution/vacua.py`). On the static-block slicing ($h_4 = 0$) the x4 constraint reads $3G(k)A^2 = \Lambda_{eff} + \kappa\rho$, with the effective-Newton factor $G(k) = 1 - 24w_2 k + 72w_3 k^2$. For every set of weights, the only static vacuum without sources is $k_0 = 1/(4w_2)$, $\Lambda_0 = -3H^2/(4w_2)$. A theorem of the homogeneous reduction: the block mode about that vacuum has $\lambda^2 = k_0 G(k_0) H^2/4$. Attractive four-dimensional gravity therefore comes with an exponentially unstable block, and a stable block comes with repulsive gravity.

   - The author's weights $(1, 1/32, 1/1152)$ give $G(k_0) = -1$: repulsive gravity.
   - The author's $\Lambda = -10H^2$ admits no static block with a small cosmological term.
   - UM-1 is therefore CHANGED to the benchmark weights $(1, 1/32, 1/768)$, which give $G(k_0) = +1$, and to $\Lambda = -24H^2 + \delta\Lambda$. A block stabiliser with $\Delta m^2 > 2H^2$ is ASSUMED; its origin is OPEN.
   - Near the vacuum the block tracks $\delta k = (\kappa p_0 - \delta\Lambda - 12A^2)/(3H^2)$. The extra times therefore keep deflating, slowly ($h_4 = 4A\dot{A}/(k_0 H^2) < 0$), and are never static.

3. Masses (section 4, DERIVED, `solution/masses.py`).

   - The photon, gluon and graviton are exactly massless.
   - The W and Z masses come from the Higgsless tip. A positive tip term gives at most $m_W = 0.01824\, m_{KK}$.
   - The Kaluza-Klein scale is fitted to the measured $\rho_0 - 1$: $m_{KK} = 121$ TeV, which gives $H = 23.4$ TeV and an x_0 period $\pi/(6H) = 4.4 \times 10^{-21}$ m.

- The electron, up, down and neutrino masses come from chiral zero modes with $c_f = f\mu_f L \approx 10$ to 19.5.
- The uncertainty estimate $\hbar c/(2\Delta s)$ exceeds every fermion mass by a factor 5×10^7 to 10^{16}. Chiral zero modes are the exception to the author's Heisenberg argument; the W mass, too, comes 99.9 % from the tip. The argument holds for the x_0 excitations, the dark matter included (`ANALYSIS.md`).

4. dirac16complex, Kohn-Sham, coupled to all fields (section 5, DERIVED, `solution/ks_coupled.py`).

   - At the vacuum slice the fold band is massless ($\varepsilon = \kappa|k|$ exactly, $g_F = 8$ states per 3-momentum).
   - The lightest odd-parity bulk level, $2.17492\,m = 50.9$ TeV, is the dark matter (a CONDITIONAL candidate: exact parity, the relic annihilation and non-universal portals are required; `ANALYSIS.md` section 8).
   - The portal mean fields of the Standard-Model fermions shift the dark-matter level linearly. They leave the fold band massless and pass the Hellmann-Feynman and y-exchange checks.
   - The coupled fixed point $\Psi \leftrightarrow$ Standard Model $\leftrightarrow$ block converges.

5. Cosmology (section 6, DERIVED, `solution/cosmology.py`).

   - UM-1 has ONE generation (the author's particle list), so its Standard-Model N_{eff} is 1.015. The Ψ fold band must supply $\Delta N_{eff} \approx 2$.
   - With $g_F = 8$ (the fold-band states of the whole cell) that requires Ψ to decouple from the plasma at $g_{*s} = 12.3$, i.e. in the QCD crossover. Decoupling above it gives $N_{eff} = 1.30$ to 1.37, excluded by about 10σ; decoupling in the pion gas gives 3.54 ($+3.2\sigma$).
   - If the four-dimensional states are counted on one half of the cell ($g_F = 4$, the other reading of the record's convention), any decoupling between neutrino decoupling and 20 MeV gives $N_{eff} = 3.015$ ($+0.15\sigma$) with no tuning, $H_0 = 67.04$ km/s/Mpc and BAO $\chi^2 = 10.9$. Which reading holds is a question of the model (OPEN).
   - With $N_{eff} = 2.99$ and θ_* fixed, $H_0 = 66.75$ km/s/Mpc and $r_{drag} = 147.4$ Mpc. The eBOSS DR16 BAO give $\chi^2 = 11.6$ for 12 points (Planck ΛCDM: 10.2).
   - The CNB has $T_\nu = (4/11)^{1/3} T_{CMB} = 1.945$ K.

6. Conflicts with data, stated plainly (section 7).

   - There is no Higgs boson, while one is observed at 125 GeV.
   - Without a Higgs, WW scattering violates perturbative unitarity near 1.2 TeV, far below m_{KK}.
   - One generation needs the tuned decoupling of item 5.
   - The cosmological term needs $\delta\Lambda/H^2 \approx 8 \times 10^{-93}$.

2. The approximation scheme

The coupled system of `FIELD_EQUATIONS.md` section 9 is solved as a series in four small parameters. Each is computed below, not assumed:

parameter	**meaning**	**size in the solution**
ϵ	metric perturbations (CMB anisotropy)	10^{-5} (STANDARD)
$(A/H)^2$	late-time expansion rate against the cell scale	3.7×10^{-93} today
$\kappa\rho/H^2$, $\kappa p_0/H^2$	sources against the cell curvature	of order $(A/H)^2$ (the Friedmann equation)
e_V, e_D, e_S	portal mean fields against m	proportional to the Standard-Model densities in units of m^3

- Order 0 in all of them: the static block vacuum (section 3), the x_0 eigenproblems of every field on the vacuum cell (section 4), and the Kohn-Sham ground state of Ψ on the vacuum cell (section 5).
- First order in $(A/H)^2$ and $\kappa\rho/H^2$: the four-dimensional Friedmann equation with $G_{eff} = G/G(k_0)$, and the quasi-static block shift that makes the extra times keep deflating (section 3.4).
- First order in the portal amplitudes: the linear response of the Kohn-Sham state of Ψ (section 5.3). It feeds back on the Standard-Model masses and on the block through the hidden pressure p_0 (section 5.5).
- First order in ϵ: the acoustic physics of the CMB and the BAO in the four-dimensional effective theory. It uses the standard fitting formulae, calibrated once on Planck ΛCDM (section 6). Its accuracy is set by the fitting formulae, a few parts in 10^4 on θ_* and r_{drag}.

3. Gravity: the Einstein-Lovelock vacuum and the block

3.1 The static-block slicing (exact)

With $\dot{a}_4 = A$, $\ddot{a}_4 = 0$, $u = 2A$ (so $h_4 = 0$ and k constant) and $\dot{u} = 0$, the record's three background components (`Revision/field_equations_a4/a4-equations.json`, generalSource) are, for general w_2, w_3 (check V1):

$$F_4 = -3G(k)A^2 + 6H^2k(1 - 2w_2k) + \Lambda + \kappa\rho, \qquad G(k) = 1 - 24w_2k + 72w_3k^2$$

$$F_0 = 6(12w_2k - 1)A^2 - 12(w_2 - 18w_3k)A^4/H^2 + 3H^2k + \Lambda - \kappa p_0$$

F_1 is F_4 with $-\kappa p_3$ in place of $\kappa\rho$. In Einstein gravity ($w_2 = w_3 = 0$), $F_4 = 0$ is the Friedmann equation $3A^2 = \Lambda + 6H^2k + \kappa\rho$. In general $A^2 = \kappa\rho/(3G) + \ldots$, so the effective four-dimensional Newton constant is $G_N/G(k)$: $1/G(k)$ is its ratio to the Einstein value. $G > 0$ means attractive gravity; $G < 0$ means a negative Newton constant, i.e. matter decelerates the expansion in the wrong direction.

3.2 The static vacuum and the effective Newton constant

$F_4 = F_0 = 0$ with $A = \rho = p_0 = 0$ has exactly one solution with $k \neq 0$ (check V2):

$$k_0 = \frac{1}{4w_2}, \qquad \Lambda_0 = -\frac{3H^2}{4w_2}, \qquad G(k_0) = -5 + \frac{9w_3}{2w_2^2}.$$

- The author's weights $w = (1, 1/32, 1/1152)$ give $k_0 = 8$, $\Lambda_0 = -24H^2$ and $G(k_0) = -1$: REPULSIVE gravity in the only static vacuum. With these weights, $G(k) > 0$ only for $k < 6 - 2\sqrt{5} = 1.528$ or $k > 6 + 2\sqrt{5} = 10.47$.
- The author's $\Lambda = -10H^2$: the hidden component gives the quasi-static $k = 10/3$, where $\Lambda_{eff} = 35H^2/6$ and $G = -29/36$. That is a cosmological term of the size of the cell curvature, with repulsive gravity. No static vacuum with a small cosmological term exists for $\Lambda = -10H^2$.

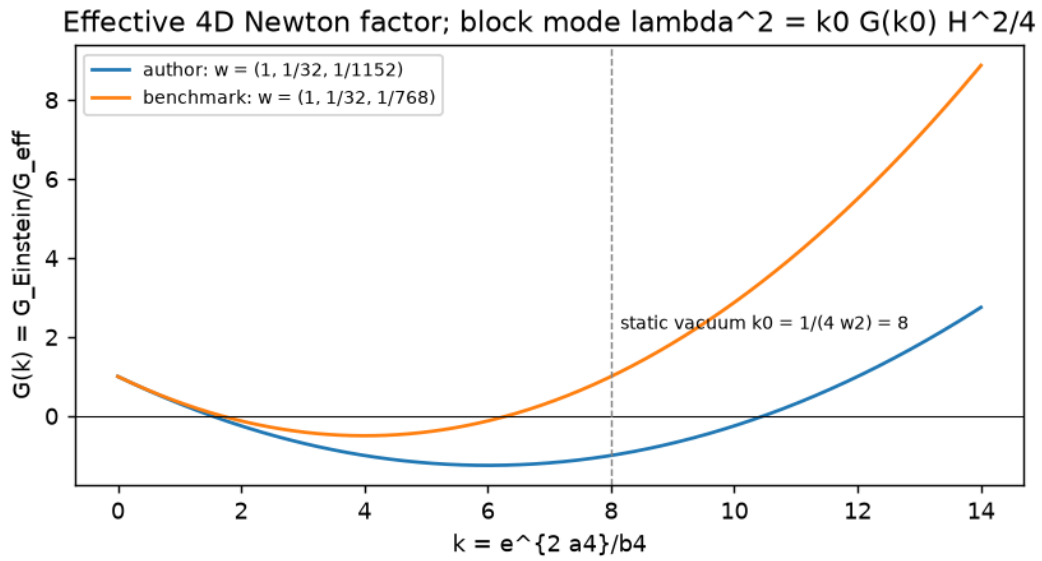


Figure 1: The effective Newton factor G(k) for the author's weights and the benchmark; the static vacuum k0 = 8.

3.3 The block mode: a theorem of the homogeneous reduction

Linearising the x_1 and x_0 components (solved for $\ddot{a}_4$ and $\dot{u}$; the Jacobian has determinant -24) together with $\dot{k} = k(2\dot{a}_4 - u)$ about the vacuum gives, for general weights (check V3), the characteristic polynomial

$$\lambda\,(32w_2^3\lambda^2 + (10w_2^2 - 9w_3)H^2) = 0, \qquad \lambda^2 = \frac{k_0\,G(k_0)}{4}\,H^2.$$

The linearised constraint vanishes identically, and the zero eigenvalue is the de Sitter direction ($u = 2\dot{a}_4$). So the block is stable (oscillating with $\omega^2 = -\lambda^2$) exactly when gravity is repulsive, and it is unstable (growing like $e^{\lambda x_4}$) exactly when gravity is attractive (check V4):

weights	$G(k_0)$	**4D gravity**	**block**
author $(1, 1/32, 1/1152)$	-1	repulsive	stable, $\omega = \sqrt{2}H$
benchmark $(1, 1/32, 1/768)$	$+1$	attractive (Einstein value)	unstable, $\lambda = \sqrt{2}H$

The instability has a natural reading in the author's picture. A static block is not where an attractive-gravity universe starts: it runs away from it at the rate $\lambda \sim H$. That is the exponential evolution of the extra times of the primordial era.

CHANGED (UM-1 version 1.1, forced by the solution):

- The cubic weight is $w_3 = 4w_2^2/3 = 1/768$, so that $G(k_0) = 1$ and the four-dimensional Newton constant has its Einstein value at the vacuum.
- $\Lambda = \Lambda_0 + \delta\Lambda = -24H^2 + \delta\Lambda$.
- A block stabiliser is ASSUMED: an additional potential for k whose curvature exceeds $k_0 G(k_0) H^2/4 = 2H^2$. The natural candidate is the renormalised Dirac-sea (Casimir) energy of Ψ and of the Standard-Model fields in the cell as a function of f, which the Kohn-Sham record does not contain (it normal-orders the sea). Its computation is OPEN.

The radion then has a mass of order H (tens of TeV). It mediates no long-range force, so the local and the cosmological Newton constants agree.

3.4 Slow deflation near the vacuum (the extra times are never static)

To first order in $\delta k = k - k_0$, $\delta\Lambda$, A^2 and the sources, $\partial F_4/\partial k = 0$ at k_0. The constraint is therefore the Friedmann equation $3G(k_0)A^2 = \delta\Lambda + \kappa\rho$, and the hidden component gives the quasi-static block

$$\delta k = \frac{\kappa p_0 - \delta\Lambda - 12A^2}{3H^2}, \qquad h_4 = -\frac{\dot{k}}{2k} = \frac{4A\dot{A}}{k_0 H^2} < 0 \quad (\dot{A} < 0).$$

The block keeps deflating as long as the universe decelerates or its Hubble rate falls: f shrinks while A falls. The rate is suppressed by $(A/H)^2$. With $\dot{A} = -(1+q)A^2$, the deflation today is $h_4/A = -4(1+q_0)(A/H)^2/k_0 \approx -10^{-93}$ (section 9). The x_0-generated masses scale like $1/f$, so they drift by the same relative amount per Hubble time, far below every bound on varying constants. This keeps the binding statement "the extra times deflate exponentially, never static" in the primordial era. It is replaced by a slow, never-vanishing deflation in the late era (a labelled departure: a late exponential deflation at a rate of order H would change the masses by order one per 10^{-28} s).

4. Masses from the x_0 wave functions

4.1 Gauge fields and the graviton

On each half of the cell, $t = y/L \in [-1, 0]$, the x_0 operator of a gauge field is $-e^{-3\eta t}(e^{3\eta t}u')' = \Lambda u$ with $\eta = HL$ and $\Lambda = (m\,fL)^2$. The fold condition is $u'(0) = 0$ and the tip (Robin) condition of the sigma field is $u'(-1) = \beta u(-1)$. Every solution is a function of $y(z) = \ln|\sin z|/(6H)$ and is therefore periodic in x_0 with period $\pi/(6H)$ (check M6).

- Unbroken generators ($\beta = 0$): $\Lambda_0 = 0$ (the photon, the gluons, and the graviton, whose zero mode is flat as well) and $\Lambda_n = 9\eta^2/4 + n^2\pi^2$ exactly (substitute $u = e^{-3\eta t/2}w$); the shooting solver reproduces these to 3×10^{-13} (check M1). $m_{KK} = \sqrt{\Lambda_1}/(fL)$.
- The Higgsless tip: a POSITIVE tip term (a positive v_Σ^2) cannot make the W heavier than the Dirichlet-tip level. That level satisfies $\tanh q = 2q/(3\eta)$, $\Lambda_D = 9\eta^2/4 - q^2 = 0.010016$ for $\eta = 3$ (check M2), so $m_W \le 0.01824\, m_{KK}$ and $m_{KK} \ge 4.41$ TeV.
- Custodial symmetry gives $c_{tip}(Z)/c_{tip}(W) = 1/\cos^2\theta_W$, so $\beta_Z = \beta_W/\cos^2\theta_W$. The saturation of $\Lambda(\beta)$ makes $\rho = m_W^2/(m_Z^2\cos^2\theta_W)$ larger than 1 by $5.6 \times 10^{-4}(100\,\text{TeV}/m_{KK})^2$. m_{KK} is FITTED to the measured $\rho_0 - 1 = 0.00038 \pm 0.00020$: $m_{KK} = 121.2$ TeV, $H = 23.42$ TeV, proper half-cell $fL = 8.9 \times 10^{-21}$ m (check M4). With m_W as input, the predicted m_Z lies 0.017 GeV below the measured value (the on-shell angle; loop corrections of a Higgsless theory are not included).
- The W mass is a tip effect. In the Rayleigh quotient of its eigenfunction the bulk hidden-momentum part is 1.3×10^{-3} of m_W^2 and the tip part is the rest (check M3).

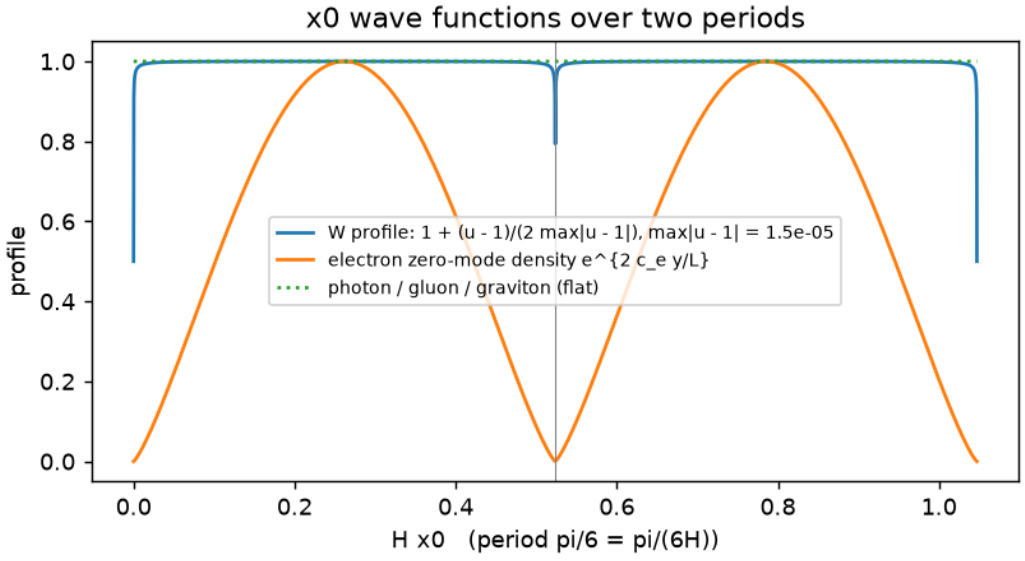


Figure 2: The x0 wave functions over two periods: the W profile, the electron zero-mode density, the flat massless modes.

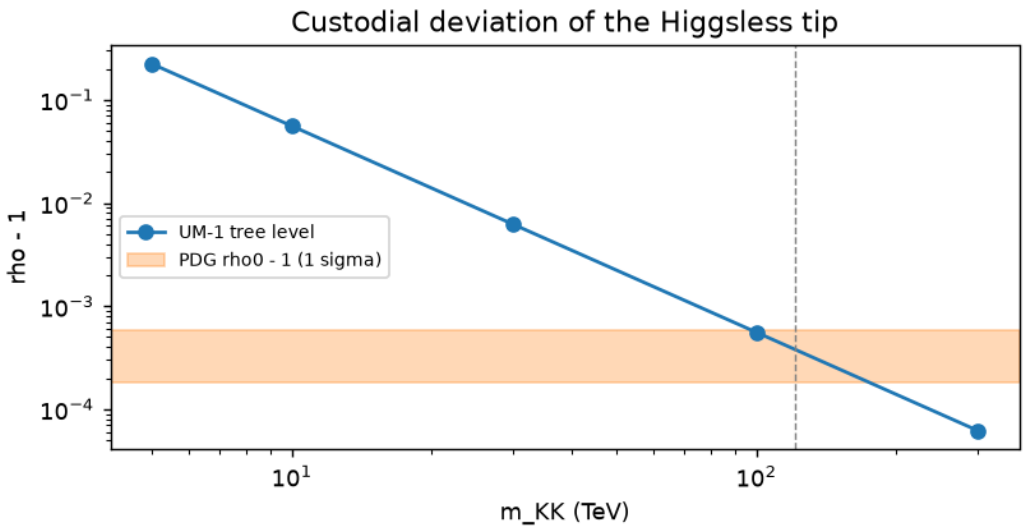


Figure 3: The tree-level custodial deviation rho - 1 against m_KK, with the measured band; the fitted m_KK.

4.2 Fermions and the Heisenberg argument

Each fermion has a chiral zero mode $\chi = (e^{ct}, 0)$ with $c = f\mu_f L$ (`FIELD_EQUATIONS.md` section 5). Its normalised tip value is $\zeta(c)^2 = 2ce^{-2c}/(1 - e^{-2c})$, and $m_f = \lambda_f m_{KK}\zeta(c_L)\zeta(c_R)$ (here $\lambda_f = 1$, $c_L = c_R = c_f$).

- m_e, m_u, m_d and $m_\nu = 0.05$ eV (one generation; chosen at $\sqrt{\Delta m^2_{atm}}$) give c_f between 10.4 and 19.5, i.e. bulk masses μ_f of 9.5 to 18.4 H (check M5; section 9).
- The profiles are localised at the fold, with proper widths $\Delta s \approx 10^{-22}$ m. The author's estimate $mc^2 \geq \hbar c/(2\Delta s)$ would give 220 to 430 TeV, while the actual masses are 2×10^{-8} to 10^{-16} of that.
- This is the DERIVED exception of `FIELD_EQUATIONS.md` (check D3 there). The hidden momentum of a chiral zero mode is cancelled by its bulk mass ($h\chi = 0$), and its mass is set by its value at the tip, not by its localisation.
- (1.2) The first-order formula is the exact x_0 eigenvalue. With the tip Yukawa as the self-adjoint mixing condition of the coupled (L, R) Dirac problem, the lowest eigenvalue equals $\kappa\zeta(c_L)\zeta(c_R)$ to 6×10^{-8} for e, u, d, ν (`GAPS.md` section 2.2).
- For bosons the Rayleigh quotient is the weighted mean $\int e^{3Hy}|\partial_s u|^2 / \int e^{3Hy}|u|^2$, the hidden momentum relative to the geometric zero mode, and not the mean squared momentum of the proper density (CORRECTED in version 1.1; `ANALYSIS.md` sections 3 to 6). The uncertainty bound therefore holds for the odd modes and the excitations ($m \geq 3H/(2f)$) and fails for the zero modes: the photon zero mode is localised within 1.4×10^{-21} m and massless. The W mass is dominated by the tip term (section 4.1), so here too the mass comes from the boundary, not from localisation.

5. dirac16complex: the Kohn-Sham solution coupled to all fields

5.1 The equations and their implementation

The Kohn-Sham block of `FIELD_EQUATIONS.md` section 6, with the portal mean fields of the Standard-Model fermions:

$$h_j = j[-(i/f)\sigma_1\partial_y + M_{eff}(y)\sigma_2 + \kappa|k|\sigma_3] + v_v(y), \qquad M_{eff} = m + \tfrac{15}{16}\lambda S + e_D P_{fold} + e_S P_{tip}, \qquad v_v = -\tfrac{1}{16}\lambda n + e_V P_{fold}$$

- The fermion profiles: $P_{fold}(y) = e^{2c_e y/L}$ is the shape of the electron zero-mode density, with $c_e = 11.20$ from section 4 (the Standard-Model fermions sit at the fold). P_{tip} is a unit-weight exponential of width 0.1 at $y = -L$, standing in for δ_{tip}.
- The portal amplitudes $e_V = g_V \sum_f q_f n_f$, $e_D = g_D \sum_f m_f s_f$ and $e_S = g_S \sum_f s_f$ (P1, P5, P2 of `MODEL.md`) are in units of m.
- The record's reference solver `Revision/kohn_sham/reference/ks_fd.py` is used unchanged, with the record's units ($H = m = 1$, $L = 3$, $dk = 0.25$, $v_t = 1$, $b_4(0) = 1$), coefficients and conventions (label sets from the free problem, both fold conditions filled).
- The portal potentials are added to the sector matrices of the state's grid (a wrapper of `sector_arrays`), and the energy-momentum tensor uses the total potentials in p_0.
- The vacuum slice $f = k_0^{-1/2} = 0.35355$ is $a_4 = 3\ln 8/2 = 3.1192$ on the record's history ($f = e^{-a_4/3}$). There $\kappa = e^{-a_4} = 0.04419$.

5.2 The spectrum of Ψ at the vacuum

Free spectrum on three grids (300, 600, 1200), Richardson-extrapolated:

- Cross-checks. The record's bulk edge at $a_4 = 0$ is reproduced to 1.5×10^{-13} (check K1). The $k = 0$ levels agree with the exact spectra of `ks-theory.json` to 10^{-14} (check K2).
- Fold zero modes: 8 states at $k = 0$ (the two even sectors).
- The fold band $\varepsilon = \kappa|k|$ holds exactly (check K3). Its quanta are MASSLESS in four dimensions, with $g_F = 8$ states per 3-momentum: 4 particles (sector $j = +1$, even) and 4 antiparticles (the holes of the sea band $\varepsilon = -\kappa|k|$ of sector $j = -1$, even). They are the dark radiation. The count is per cell, i.e. both halves of a period of x_0 (the record's particle-number convention, `ks-theory.json` densities: each half holds half of the particles); counted on one half it is $g_F = 4$ (section 6).
- The lightest odd-parity level is $\varepsilon = 2.174920\,m$ at $k = 0$; the lightest even bulk level is $3.12618\,m$. The fold reflection $z \to \pi - z$ maps the even and odd sectors to themselves. If this parity is exact (ASSUMED; the self-interaction and the portals are reflection symmetric), the lightest odd state is stable: the DARK MATTER, with $M_{DM} = 2.17492\,m = 2.17492\,H = 50.9$ TeV ($m = H$ in the record's units).

- Odd quanta annihilate in pairs into fold-band quanta through the self-interaction λ, so the relic density is set within the dark sector. CORRECTED in version 1.1 (`ANALYSIS.md` section 8.4): with the light states of UM-1 ($g_* = 60.25$, $x_f = 29.5$) a Dirac relic of 50.9 TeV needs $\langle\sigma v\rangle = 6.5 \times 10^{-26}$ cm^3/s (the familiar 4.4×10^{-26} assumes the Standard-Model $g_* = 106.75$). That is 37 % of the thermal s-wave unitarity limit, a contact strength $G \approx 1.4/H^2$. The reduction of λ to this cross-section is OPEN.

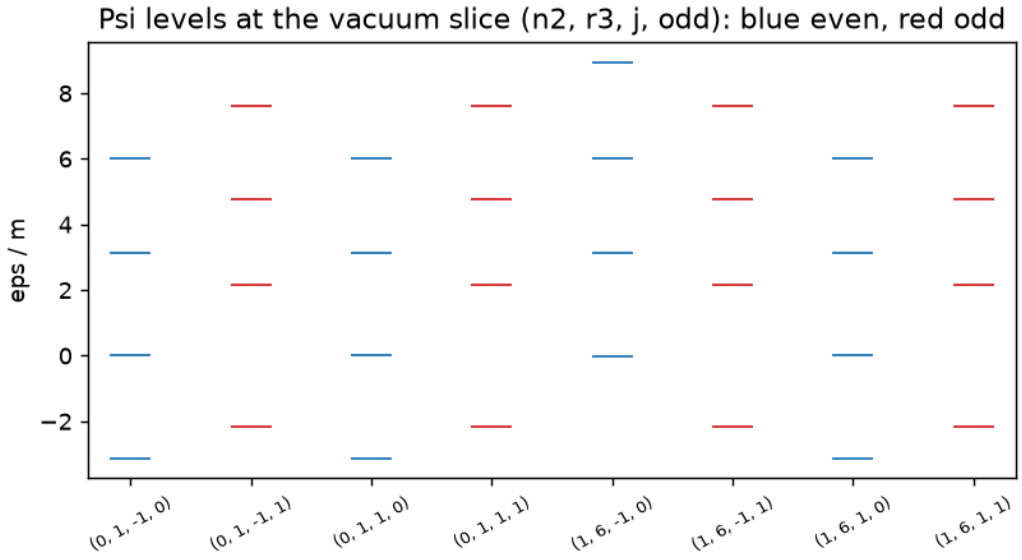


Figure 4: Levels of Psi at the vacuum slice by sector (n2, r3, j, odd).

5.3 Coupling to the Standard Model: linear response

Ground states $N = 8$ (the fold zero modes) and $N = 592$ (a fold-band gas) are computed for $\lambda = 0$ and for λ_1 of the record (157.7 and 2.131). For each portal the solver gives the following (table in section 9):

- the Hellmann-Feynman check: dE_{KS}/de from Richardson differences at fixed occupations equals $\int(\partial M/\partial e\, S + \partial v/\partial e\, n)$ (check K4). The scalar portals do not act on the chiral zero modes at $\lambda = 0$, whose scalar density vanishes; these are null cases.
- the y-exchange identity at $e = 0.02$ (check K5), the conservation law of the hidden direction with the force of the external field:

$$[e^{3Hy} p_0]_{-L}^{0} - 3H \int e^{3Hy} p_t \, dy = - \int e^{3Hy} (M'_{ext} S + v'_{ext} n) \, dy,$$

satisfied to $O(h^2)$.

- the response of the dark-matter level, of the fold band and of the sources.

The fold band stays massless. Without the self-interaction its gap at the lowest shell changes by less than 10^{-14} per unit e, and the vector portal shifts all fold levels together (a chemical-potential shift). With λ_1 the vector portal changes it by -4.0×10^{-7} per unit e, a second-order mean-field effect of no consequence for portal amplitudes of physical size. The dark-matter level moves linearly with every portal; this is the dark-matter–Standard-Model interaction of the model in the mean-field limit.

5.4 Cold dark matter

Eight dark-matter quanta are placed in the lowest odd levels ($k = 0$, $j = \pm 1$) on top of the $N = 8$ ground state, with fixed occupations and solved self-consistently on three grids. The energy they add, integrated from the energy-momentum tensor over the cell, is 8 times the analytic odd level to 10^{-9} (check K6, the bookkeeping of the fixed-occupation states). At $\lambda = 0$ the added matter has $w = p_3/\rho = 0$ by construction ($k = 0$ and no interaction energy); with λ_1 it has $w = 0.0018$. These are $k = 0$ eigenstates, not a relic momentum distribution. The quanta have a hidden pressure $p_0/\rho = 0.66$ (section 9), which pushes on the block through $\delta k = \kappa p_0/(3H^2)$.

5.5 The coupled fixed point

The loop runs as follows:

1. the Kohn-Sham state with the vector portal of the electron density;
2. its x0-averaged hidden pressure $\bar{p}_0$;
3. the block shift $\delta k = \sigma \bar{p}_0/3$, with $\sigma = \kappa/H^2$ in the units of the Kohn-Sham densities;
4. the new $f = (k_0 + \delta k)^{-1/2}$ and the new $c_f = f\mu_f L$ and $m_{KK} \propto 1/f$;
5. the new portal profile and the new Standard-Model masses;
6. the Kohn-Sham state again.

The loop stops when successive block shifts agree to 10^{-10} relative (check K7). For $\sigma = 0.1$ the map is nearly constant (the loop is in effect one evaluation); for $\sigma = 10^4$ and 10^6 the hidden pressure responds to the block shift and the loop is a genuine fixed-point iteration, whose contraction factor is listed in section 9 with the shifts of f and of the masses. Only the hidden pressure of Ψ sources the block here; the Standard-Model share of the portal stress is left out (it is of the same order as the portal energy, small at these amplitudes). At late times $\sigma\bar{p}_0$ is of order $(A/H)^2$, so the feedback is of order 10^{-93}. The coupling of Ψ to all other fields matters in the primordial era, which is the subject of the record's phases W5 to W8.

6. CMB, CNB and BAO

The late universe is the four-dimensional effective theory of the vacuum with $G_{eff} = G$ (benchmark). Its components are:

- photons ($T_{CMB} = 2.7255$ K);
- one neutrino species ($N_{eff,SM} = 3.044/3 = 1.0147$, $m_\nu = 0.05$ eV);
- the Ψ fold band (massless, $g_F = 8$, decoupled at T_{dec});
- the Ψ dark matter ($\omega_c = 0.1200$, set by the relic density);
- baryons from u and d by QCD ($\omega_b = 0.02237$);
- $\delta\Lambda$, for flatness.
- $\Delta N_{eff} = (g_F/2)(T_\Psi/T_\nu)^4$ with $(T_\Psi/T_\nu)^3 = 7.25/g_{*s}(T_{dec})$. Here g_{*s} counts the UM-1 content: one generation and no Higgs, so it is 53.25 above m_W, 44.25 down to the QCD crossover, 10.25 in the pion gas and 7.25 below 20 MeV.
- Fitting formulae: Hu-Sugiyama z_* and Eisenstein-Hu z_{drag}; the sound horizon is a numerical integral. They are calibrated once on Planck 2018 ΛCDM (the factors 1.00198 on θ_* and 0.97684 on r_{drag}). h is fitted to $100\theta_* = 1.04110$.
- The first peak uses $\ell_1 = \ell_A(1 - \phi_1)$ with $\phi_1 = 0.267(r_*/0.3)^{0.1}$.
- BAO: the eBOSS DR16 BAO-only consensus, 12 measurements from $z = 0.15$ to 2.33.

Results (section 9):

- Decoupling above the QCD crossover gives $N_{eff} = 1.30$ to 1.37, excluded at about 10σ. It also forces $H_0 \approx 48$ km/s/Mpc at fixed θ_* and ω's, and BAO $\chi^2 > 600$.
- Pion-gas decoupling gives $N_{eff} = 3.54$ ($+3.2\sigma$, BAO $\chi^2 = 33.8$). Decoupling below 20 MeV gives 5.01.
- $N_{eff} = 2.99$ requires $g_{*s}(T_{dec}) = 12.31$: Ψ must decouple inside the QCD crossover (about 150 MeV). Then $H_0 = 66.75$ km/s/Mpc, $r_{drag} = 147.39$ Mpc, $\ell_A = 301.76$ (fixed by the fit of θ_*), $\ell_1 = 220.7$, and BAO $\chi^2 = 11.6$ for 12 points.
- This is the model's sharpest cosmological prediction: the dirac16complex fold band must stand in for the two missing neutrino species. With the whole-cell count ($g_F = 8$) its decoupling temperature is fixed to the QCD crossover, which with the 4D portal scale sets $G_V^{-1/2} \approx 13$ TeV (a neutrino-like decoupling estimate, STANDARD).
- With the one-half count ($g_F = 4$) decoupling anywhere between about 2 and 20 MeV gives $N_{eff} = 3.015$ ($+0.15\sigma$), $H_0 = 67.04$ km/s/Mpc, $r_{drag} = 147.27$ Mpc and BAO $\chi^2 = 10.9$: a fit as good as ΛCDM without tuning. The counting convention (whether the four-dimensional observer's states are those of the whole cell or of one half) decides between the two; it is OPEN.
- (1.2) The dark-matter density $\omega_c = 0.1200$ is a Planck input here. In UM-1 version 1.3 (P1 on the lepton multiplets) the portal that gives this decoupling also fixes the relic abundance: $\Omega_{DM}h^2 = 0.115$ at $T_{dec} = 150$ MeV, 0.06 to 0.20 over the crossover, and 0.120 at 152 MeV (`GAPS.md` section 3). The input is therefore consistent with the model's own prediction.
- CNB: $T_\nu = 1.9454$ K and $n_\nu = 112$ cm^{-3} for the one species. The Ψ fold background has $T_\Psi = 1.63$ K and $n_\Psi = 264$ cm^{-3} (section 9). Both are relativistic today except the massive ν ($m_\nu \gg T_\nu$).

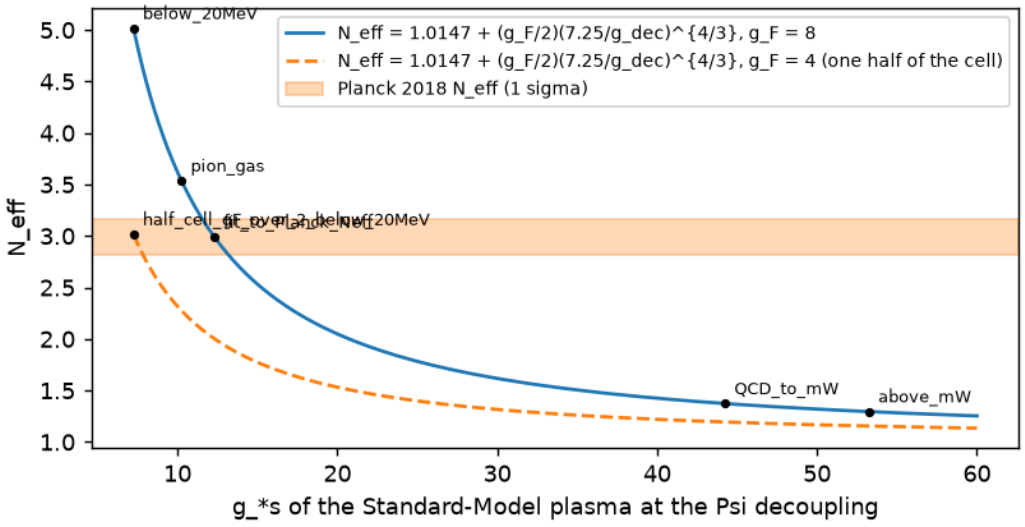


Figure 5: N_eff against g_*s at the Psi decoupling, with the Planck band and the scenarios.

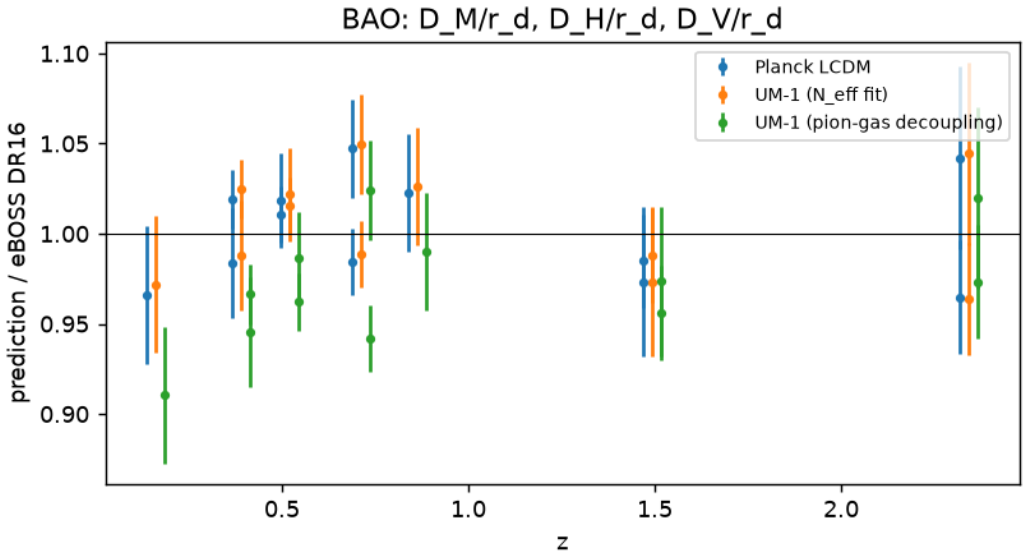


Figure 6: BAO predictions over eBOSS DR16 data for Planck LCDM and two UM-1 scenarios.

7. Assumptions, changes, conflicts and open problems

- CHANGED: $w_3 = 1/768$ (author: $1/1152$) and $\Lambda = -24H^2 + \delta\Lambda$ (author: $-10H^2$). The author's values give repulsive four-dimensional gravity in the only static vacuum (section 3.2).
- ASSUMED:
  - a block stabiliser with curvature above $2H^2$ (section 3.3);
  - the exact fold-reflection parity that keeps the dark matter stable (exact to 2×10^{-27} in amplitude for a lifetime above 10^{26} s; `ANALYSIS.md` section 8.3);
  - $m = H$ for Ψ (the record's units);
  - $\lambda_f = 1$ and $c_L = c_R$ for the fermions;
  - $m_\nu = 0.05$ eV;
  - the fermion profile shapes of the portal fields;
  - Planck's ω_b and ω_c as inputs;
  - the whole-cell count of the fold-band states ($g_F = 8$; the one-half count is shown as the alternative).
- LABELLED DEPARTURE: in the late era the extra times deflate at the rate $h_4 \approx -10^{-93}A$ (section 3.4), not exponentially. They are never static.
- CONFLICTS WITH DATA:
  - No Higgs, while a 125 GeV Higgs boson is observed. Without it, the longitudinal WW amplitude grows like s/v^2 and violates unitarity near $\sqrt{8\pi}\,v \approx 1.2$ TeV, while the KK modes that could restore it lie at 121 TeV.
  - The S parameter of the Higgsless tip is not computed (OPEN).
  - One generation makes N_{eff} fit only for a decoupling inside the QCD crossover (whole-cell count, $g_F = 8$), or naturally with the one-half count ($g_F = 4$, section 6).
  - The cosmological term needs $\delta\Lambda/H^2 \approx 7.5 \times 10^{-93}$ (the cosmological-constant problem, unsolved here). No state of Ψ can stand in for it (`ANALYSIS.md` section 9).
  - (1.1) If the vector portal that gives $N_{eff} = 2.99$ acts on the dark matter with its x_0 overlap, the dark matter is under-abundant ($\Omega h^2 = 0.023$) and σ_{SI} is 1.6 times the LZ limit (`ANALYSIS.md` section 8.5): the portal coefficients must differ between Ψ modes, or the one-half count must hold. (1.2) Resolved in UM-1 version 1.3 by the lepton-only P1: $\Omega h^2 = 0.115$ predicted, $\sigma_{SI} = 8 \times 10^{-50}$ cm^2 (`GAPS.md` section 3).
- OPEN:
  - the stabiliser (the Casimir energy of the cell);
  - the reduction of the Psi self-interaction to the relic cross-section;
  - the portal coefficients;
  - the primordial-era solution (the W phases of the Revision record);
  - the order-ϵ Lovelock corrections to the perturbations (only the effective Newton constant is used);
  - BBN with $N_{eff}(T)$;
  - the CMB damping tail, which needs a recombination code.

8. Reproduction

From `Revision/universe_model/solution` (every calculation under `timeout -k 5 423`):

```
python ks_coupled.py spectrum         # 4 s
python ks_coupled.py portal8_lam0     # 50 s
python ks_coupled.py portal8_lam1     # 141 s
python ks_coupled.py portal592_lam0   # 53 s
python ks_coupled.py portal592_lam1   # 158 s
python ks_coupled.py dm_lam0          # 60 s
python ks_coupled.py dm_lam1          # 107 s
python ks_coupled.py loop             # 190 s
python solve_model.py                 # 35 s: vacua, masses, cosmology,
                                      # figures, checks, the tables below
```

The PDF is built from the repository root with `python Revision/universe_model/tools/build_pdfs.py SOLUTION`; the gate steps are um1-* in `Revision/verify_revision.{sh,ps1}`.

9. Results

9.1 Generated tables (from `solution/solve_model.py`; every number below is from the run)

Vacuum and block (vacua.py).

weights w	k_0	Λ_0/H^2	$G(k_0)$	**4D gravity**	**block mode** λ^2/H^2
(1, 1/32, 1/1152)	8	-24	-1	repulsive	-2 (stable (oscillatory))
(1, 1/32, 1/768)	8	-24	1	attractive	2 (unstable (real eigenvalue))

Mass spectrum (masses.py; $\eta = HL = 3$, $f = k_0^{-1/2}$).

quantity	**value**
m_{KK} (first photon/gluon/graviton KK level, fitted to $\rho_0 - 1$)	1.212×10^5 GeV
H	23421 GeV
proper half-cell fL	8.936×10^{-21} m
period of x_0, $\pi/(6H)$	2.236×10^{-5} GeV^{-1} = 4.411×10^{-21} m
largest m_W of a positive tip term	$0.01824\, m_{KK}$ (so $m_{KK} \geq 4407$ GeV)

β_W, β_Z (tip Robin parameters)	0.01194, 0.01537
m_Z predicted (from m_W, on-shell $\cos^2\theta_W$)	91.17068 GeV (observed 91.188; difference -0.01732 GeV)
$\rho - 1$ (tree)	3.800×10^{-4}
W: bulk hidden-momentum part / tip part of m_W^2	0.001322 / 0.9987
first W KK level	1.212×10^5 GeV
photon, gluon, graviton	$\Lambda_0 = 0$: massless

fermion	m_f **(GeV)**	$m_f/m_{KK} = \zeta(c_f)^2$	$c_f = f\mu_f L$	μ_f/H
nu	5.000×10^{-11}	4.126×10^{-16}	19.54	18.43
e	5.110×10^{-4}	4.217×10^{-9}	11.2	10.56
u	0.00216	1.782×10^{-8}	10.44	9.844
d	0.0047	3.878×10^{-8}	10.03	9.458

fermion	**width** Δt	Δs **(m)**	$\hbar c/(2\Delta s)$ **(GeV)**	m_f **/ Heisenberg estimate**
nu	0.02558	2.286×10^{-22}	4.316×10^5	1.159×10^{-16}
e	0.04466	3.991×10^{-22}	2.472×10^5	2.067×10^{-9}
u	0.04789	4.280×10^{-22}	2.305×10^5	9.369×10^{-9}
d	0.04984	4.454×10^{-22}	2.215×10^5	2.122×10^{-8}

dirac16complex, Kohn-Sham (ks_coupled.py; units $m = H$).

quantity	**value**
vacuum slice $a_4 = 3\ln 8/2$, f, $\kappa = e^{-a_4}$	3.119, 0.3536, 0.04419
fold zero modes; fold band	8 at $k = 0$; $\varepsilon = \kappa\|k\|$ exactly (massless), $g_F = 8$
lowest odd bulk level (dark matter)	$2.17492\,m = 50939$ GeV
lowest even bulk level	$3.126177\,m$
dark matter ($\lambda = 0$): energy per 8 quanta, $w = p_3/\rho$, p_0/ρ	17.39936, 0, 0.6575
dark matter ($\lambda = 157.7$): energy per 8 quanta, $w = p_3/\rho$, p_0/ρ	17.4307, 0.001817, 0.6567

state (grid 600)	**portal**	dE_{KS}/de	**Hellmann-Feynman deviation**	**y-exchange residual ($e = 0.02$)**
N = 8, λ = 0	eD	0	null (both zero)	null (all terms zero)
N = 8, λ = 0	eS	0	null (both zero)	null (all terms zero)
N = 8, λ = 0	eV	0.7865	2.259×10^{-14}	7.609×10^{-5}
N = 8, λ = 157.7	eD	4.689×10^{-5}	3.690×10^{-10}	1.700×10^{-6}
N = 8, λ = 157.7	eS	4.656×10^{-4}	5.677×10^{-11}	2.986×10^{-7}
N = 8, λ = 157.7	eV	0.7865	2.856×10^{-13}	7.149×10^{-5}
N = 592, $\lambda = 0$	eD	0	null (both zero)	null (all terms zero)
N = 592, $\lambda = 0$	eS	0	null (both zero)	null (all terms zero)
N = 592, $\lambda = 0$	eV	58.2	1.343×10^{-15}	7.609×10^{-5}
N = 592, $\lambda = 2.131$	eD	0.00347	8.071×10^{-10}	1.701×10^{-6}
N = 592, $\lambda = 2.131$	eS	0.03446	1.731×10^{-10}	2.985×10^{-7}
N = 592, $\lambda = 2.131$	eV	58.2	2.998×10^{-13}	7.149×10^{-5}

state (grid 600)	**portal**	$d\varepsilon_{DM}/de$	$d(\varepsilon_{n2=1} - \varepsilon_0)/de$	$d\bar{p}_0/de$
N = 8, $\lambda = 0$	eD	0.006409	below 10^{-14}	0
N = 8, $\lambda = 0$	eS	0.03543	below 10^{-14}	0
N = 8, $\lambda = 0$	eV	0.03231	below 10^{-14}	-2.564×10^{-7}
N = 8, $\lambda = 157.7$	eD	0.006414	8.262×10^{-11}	3.432×10^{-11}
N = 8, $\lambda = 157.7$	eS	0.03571	1.947×10^{-9}	6.440×10^{-10}
N = 8, $\lambda = 157.7$	eV	0.03266	-3.968×10^{-7}	-2.564×10^{-7}
N = 592, $\lambda = 0$	eD	0.006409	below 10^{-14}	0
N = 592, $\lambda = 0$	eS	0.03543	below 10^{-14}	0
N = 592, $\lambda = 0$	eV	0.03231	below 10^{-14}	-1.898×10^{-5}
N = 592, $\lambda = 2.131$	eD	0.006414	8.262×10^{-11}	2.540×10^{-9}

N = 592, λ = 2.131	eS	0.03571	1.947×10^{-9}	4.766×10^{-8}
N = 592, λ = 2.131	eV	0.03266	-3.968×10^{-7}	-1.898×10^{-5}

coupled fixed point	**iterations**	**contraction**	δk	$\delta f/f$	$\delta m_e/m_e$	$\delta m_\nu/m_\nu$
$\sigma = 0.1$	3	0	-1.617×10^{-8}	1.011×10^{-9}	-2.263×10^{-8}	-3.950×10^{-8}
$\sigma = 10000$	5	7.472×10^{-4}	-0.001618	1.011×10^{-4}	-0.002262	-0.003946
$\sigma = 1.000$ \times 10^{6}	11	0.08263	-0.1755	0.01115	-0.221	-0.3534

Cosmology (cosmology.py).

model	g_{*s} **at decoupling**	N_{eff}	**pull vs Planck**	H_0 **(km/s/Mpc)**	**BAO** χ^2 **(12 points)**
Planck LCDM (reference)	-	3.046	-	67.36	10.25
UM-1, above mW	53.25	1.295	-9.97	47.33	680.7
UM-1, QCD to mW	44.25	1.373	-9.51	48.22	614
UM-1, pion gas	10.25	3.536	3.21	73.11	33.77
UM-1, below 20MeV	7.25	5.015	11.9	90.63	390
UM-1, fit to Planck Neff	12.31	2.99	0	66.75	11.64
UM-1, half cell gF over 2 below 20MeV	7.25	3.015	0.145	67.04	10.86

model	$100\theta_*$	z_*	r_{drag} **(Mpc)**	ℓ_A	ℓ_1
Planck LCDM (reference)	1.0411	1091.85	147.09	301.76	220.7
UM-1, above mW	1.0411	1091.85	156.69	301.76	222.8
UM-1, QCD to mW	1.0411	1091.85	156.21	301.76	222.7

UM-1, pion gas	1.0411	1091.85	144.79	301.76	220.2
UM-1, below 20MeV	1.0411	1091.85	138.48	301.76	218.7
UM-1, fit to Planck Neff	1.0411	1091.85	147.39	301.76	220.7
UM-1, half cell gF over 2 below 20MeV	1.0411	1091.85	147.27	301.76	220.7

background today	**value**
$T_\nu = (4/11)^{1/3} T_{CMB}$	1.9454 K
n_ν (one species, $\nu + \bar{\nu}$)	112 cm^{-3}
T_Ψ of the fold band (N_eff fit)	1.631 K
n_Ψ of the fold band ($g_F = 8$)	264 cm^{-3}
$(H_0/H)^2$	3.696×10^{-93}
deflation today $h_4/\dot{a}_4$	-8.857×10^{-94}
$\delta\Lambda/H^2$ (dark energy)	7.545×10^{-93}

Checks (`solution/reports/solution-checks.json`, 21 of 21 PASS): V1, V2, V3, V4, M1, M2, M3, M4, M5, M6, M7, K1, K2, K3, K4, K5, K6, K7, C1, C2, C3.